

Example

$$\begin{cases} 2x + 4y + 6z + 8w = 1 \\ 3x + \quad \quad \quad 7z + \quad \quad w = 10 \end{cases}$$

$\frac{1}{2}(\text{Eq 1})$

$$\begin{cases} x + 2y + 3z + 4w = \frac{1}{2} \\ 3x + \quad \quad \quad 7z + \quad \quad w = 10 \end{cases}$$

$$(\text{Eq 2}) - (3\text{Eq 1}) \rightarrow \begin{cases} x + 2y + 3z + 4w = \frac{1}{2} \\ -6y - 2z - 11w = \frac{17}{2} \text{ etc.} \end{cases}$$

Matrix

$\xrightarrow{\text{notation}}$
 $A\vec{v} = \vec{b}$

$$\begin{bmatrix} 2 & 4 & 6 & 8 \\ 3 & 0 & 7 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 1 \\ 10 \end{bmatrix}$$

$\left\{ \begin{array}{l} \text{Augmented matrix} \end{array} \right.$

$[A|b]$

$$\left[\begin{array}{cccc|c} 2 & 4 & 6 & 8 & 1 \\ 3 & 0 & 7 & 1 & 10 \end{array} \right]$$

rows $\leftrightarrow$ eqⁿs

columns (except last) $\leftrightarrow$ variables

Observe, doing the following gives an equivalent system of equations

① Replace Eq_i^n by $\text{Eq}_i^n + a(\text{Eq}_j^n)$ for $a \in \mathbb{R}$

② $\text{Eq}_i^n \leftarrow c(\text{Eq}_i^n)$ where $c \neq 0$

③ Swap Eq_i^n & Eq_j^n .

Associated elementary row operations on augmented matrix

① $(i^{\text{th}} \text{ row}) + a(j^{\text{th}} \text{ row})$ " $R_i + aR_j$ "

② $c(j^{\text{th}} \text{ row})$ " cR_i "

③ Swap i^{th} & j^{th} rows " $R_i \leftrightarrow R_j$ "

$$\left[\begin{array}{cccc|c} 2 & 4 & 6 & 8 & 1 \\ 3 & 0 & 7 & 1 & 10 \end{array} \right] \xrightarrow{\frac{1}{2}R_1} \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & \frac{1}{2} \\ 3 & 0 & 7 & 1 & 10 \end{array} \right]$$

$$\xrightarrow{R_2 - 3R_1} \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & \frac{1}{2} \\ 0 & -6 & -2 & -11 & \frac{17}{2} \end{array} \right] \xrightarrow{-\frac{1}{6}R_2} \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & \frac{1}{2} \\ 0 & 1 & \frac{1}{3} & \frac{11}{6} & -\frac{17}{12} \end{array} \right]$$

$$\xrightarrow{R_1 - 2R_2} \left[\begin{array}{cccc|c} 1 & 0 & \frac{7}{3} & \frac{1}{3} & \frac{20}{6} \\ 0 & 1 & \frac{1}{3} & \frac{11}{6} & -\frac{17}{12} \end{array} \right]$$

- pivot = leading nonzero entry in a row → usually used only when matrix is in row echelon form
- A matrix is said to be in row echelon form (REF) if "pivots in successive rows move to the right" i.e. $\forall$ rows i (except last)

$$(\text{column of leading nonzero entry in row } i+1) > (\text{column of leading nonzero entry in row } i)$$
- Typically we scale to make pivot entry = 1.

Ex. Any matrix can be converted to REF by a sequence of elem row opⁿ

Swap to get leftmost nonzero entry into top row. Kill all entries below it by ① { Remove 1st row
Continue!